DAY 06 · BEGINNER · FREE API

Memory

Short-term and long-term are two different things. Today you build both, and find out what remembering costs.

InMemorySaver | thread_id | trim_messages | InMemoryStore | SqliteSaver

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TEN DAYS, THREE WEEKS, ONE NOTEBOOK A DAY

The 10-Day Course Map

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WEEK 1 TALKING TO A MODEL

01 Your First AI Program
Ask a model, print the answer

02 Roles and Messages
The system message sets the job

03 Prompt Templates
One template, many inputs

YOU WILL HAVE BUILT
a script that answers anything you type into it

WEEK 2 BUILDING SOMETHING

04 Chains: The Pipe
`prompt | model | parser`

05 Structured Output
An object, not a paragraph

YOU ARE HERE
06 Memory
A checkpointer and a thread id

YOU WILL HAVE BUILT
a chatbot that remembers what you said a minute ago

WEEK 3 GIVING IT ABILITIES

07 Tools
Your Python, called by the model

08 Your First Agent
It decides when to call them

09 Chunks and Vectors
Find a paragraph by meaning

10 RAG with RAGWire
Citations, and a refusal

One notebook and one deck a day, **60 to 90 minutes** each. Every model used is a **free** one, and Docker only turns up on Day 9.

How Short-Term Memory Works

TURN 1 SENDS

"My name is Priya"

1 message

TURN 2 SENDS

"My name is Priya"

"Nice to meet you"

"Book a table"

3 messages

TURN 3 SENDS

"My name is Priya"

"Nice to meet you"

"Book a table"

"Friday at 7?"

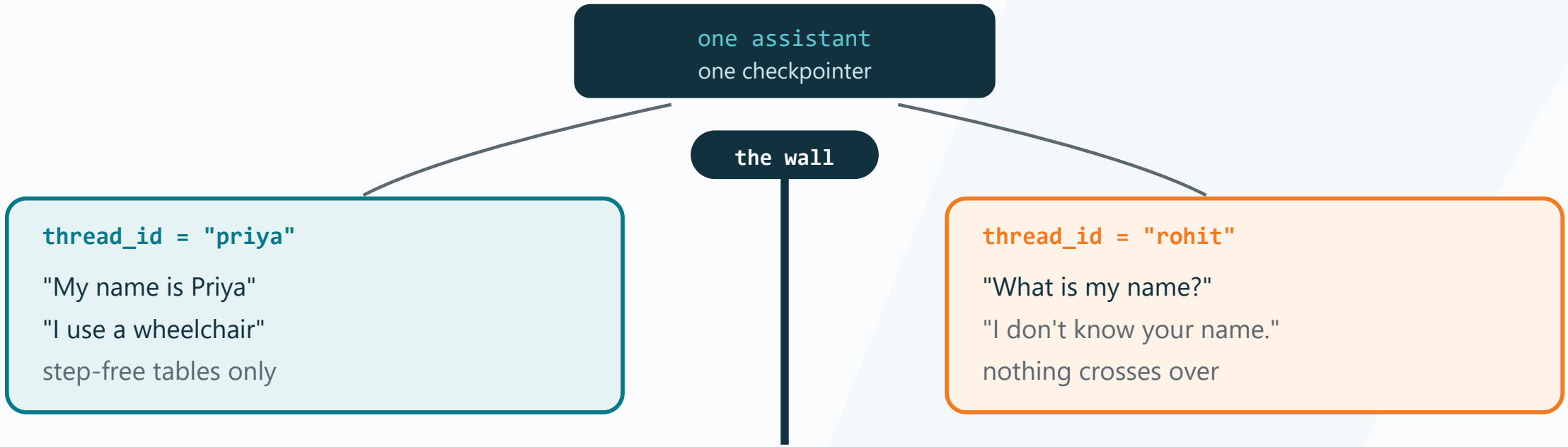
"Yes"

5 messages

There is no memory inside the model. **You resend everything, every turn.**

Short-term memory is not something the model has. It is **something you send**.

thread_id Keeps Users Separate



The wall between two people's conversations is **a string you chose**.

Shortening Memory: Trim or Summarise

TRIM

"My name is Priya" dropped

"Nice to meet you" dropped

"For four people" kept

"Around eight" kept

Free. Exact wording kept.

Loses the start of the conversation.

SUMMARISE

"Priya, wheelchair user, wants a table on Friday for four."

"Around eight" kept exactly

Early facts survive, in some form.

Loses exact wording, and detail.

costs one extra model call

Choosing is a decision about **what your application cannot afford to forget**.

Short-Term vs Long-Term Memory

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SHORT-TERM

What was said in this conversation

"a table on Friday, for four, at eight"

lives as long as the conversation

`checkpointer + thread_id`

should expire

LONG-TERM

What is true about this person, always

"Priya uses a wheelchair. Priya is vegetarian."

still true in March

`a store, read into the prompt`

should not expire

Mix these up and you get an assistant that **forgets your name every morning**.

Tomorrow: Tools


```
assistant = create_agent(model=model, tools=[], checkpointer=...)
```

it has been sitting there all day

A live rate

frozen at training time,
and it knows it

Your documents

never seen a single
file of yours

Your code

cannot run a line
of your Python

Tomorrow you hand it a Python function, and run the loop **by hand**.